

1 Theorem 10.2

Let (X, d) be a compact metric space. Then, every sequence in (X, d) has a convergent subsequence.

Proof. Let (X, d) be compact and let $\{x_n\}$ be any sequence in X .

Case 1: $\{x_n\}$ has only finitely many values.

Then, by the pigeonhole principle, there exists $x \in X$ such that $x_n = x$ infinitely often. This gives a subsequence $\{x_{n_k}\}$ that converges to x .

Case 2: $\{x_n\}$ has infinitely many values.

By Lemma 10.4, $\{x_n\}$ has a limit point $x \in X$. Since x is a limit point in $\{x_n\}$,

- There exists $n_1 \in \mathbb{N}$ such that $d(x_{n_1}, x) < 1$
- There exists $n_2 \in \mathbb{N}$ such that $d(x_{n_2}, x) < \frac{1}{2}$ where $n_2 > n_1$ and $x_{n_1} \neq x$.
- For all $k \in \mathbb{N} \setminus \{1\}$, there exists $n_k \in \mathbb{N}$ such that $d(x_{n_k}, x) < \frac{1}{k}$ where $n_k > n_{k-1}$ using $x_{n_k} \neq x$.

By construction, $\{x_{n_k}\}$ is a subsequence of $\{x_n\}$ and x_{n_k} converges to x . $\square$

2 Lemma 10.4

Let (X, d) be compact and let $E \subset X$ be infinite. Then E has a limit point in X .

Proof by Contradiction. Suppose for contradiction that E has no limit point in X . Then, for all $x \in X$, x is not a limit point of E . This means that we can find $r_x > 0$ such that $B_{r_x}(x)$ contains no points of E besides maybe x . Therefore, each $B_{r_x}(x)$ contains at most one point of E . Since this is for all $x \in X$, $\{B_{r_x}(x)\}_{x \in X}$ is an open cover of X . Since X is compact, $\{B_{r_x}(x)\}_{x \in X}$ has a finite subcover containing at most one element of X each and thereby at most one element of E each. Therefore, E is finite. This contradicts that the cardinality of E is infinite. $\square$

3 Theorem 10.8

- (a) Every compact metric space is complete
- (b) $\mathbb{R}$ is complete

Proof (a). A complete metric has every Cauchy sequence converge. Let (X, d) be compact and let $\{x_n\}$ be cauchy. By Theorem 10.2, $\{x_n\}$ has a convergent subsequence. By Lemma 10.9, $\{x_n\}$ converges. Therefore, X is complete. $\square$

Proof (b). In $\mathbb{R}$, every convergent sequence $\{x_n\}$ is bounded, meaning it is contained in some $[a, b]$. $[a, b]$ is compact and $\{x_n\}$ is bounded, so it has a convergent subsequence and converges, so $\mathbb{R}$ is complete. $\square$

4 Lemma 10.9

Let $\{x_n\}$ be cauchy. If $\{x_n\}$ has a convergent subsequence, it converges.

Proof. Suppose that $\{x_{n_\ell}\}$ is a subsequence of $\{x_n\}$ that converges to x . Therefore, given $\varepsilon > 0$, there exists $K \in \mathbb{N}$ such that if $k \geq K$, $d(x_{n_k}, x) < \frac{\varepsilon}{2}$. Similarly, since $\{x_n\}$ is cauchy, there exists some $N \in \mathbb{N}$ such that if $m \geq n \geq N$, $d(x_m, x_n) < \frac{\varepsilon}{2}$. Pick K big enough that $n_K \geq N$. This means that if $m \geq n_K$, then $\frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon > d(x_{n_k}, x) + d(x_m, x_{n_k}) \geq d(x_m, x)$. Therefore, $\{x_n\}$ converges to x . $\square$